In [1]:
```
from sklearn.datasets import load_breast_cancer
cancer = load_breast_cancer()
```

In [2]:
```
cancer.keys()
```

Out[2]:
```
dict_keys(['data', 'target', 'frame', 'target_names', 'DESCR', 'feature_nam
es', 'filename', 'data_module'])
```

In [3]:
```
cancer['data'].shape
```

Out[3]:
```
(569, 30)
```

In [4]:
```
X = cancer['data']
y = cancer['target']
```

In [5]:
```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y)
```

In [6]:
```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
scaler.fit(X_train)
```

Out[6]:
```
▾ StandardScaler
StandardScaler()
```

In [8]:
```
X_train = scaler.transform(X_train)
X_test = scaler.transform(X_test)
```

In [9]:
```
from sklearn.neural_network import MLPClassifier
```

In [11]:
```
mlp = MLPClassifier(hidden_layer_sizes=(30,30,30))
```

In [12]:
```
mlp.fit(X_train,y_train)
```

```
C:\Users\MGM\anaconda3\Lib\site-packages\sklearn\neural_network\_multilayer_
perceptron.py:690: ConvergenceWarning: Stochastic Optimizer: Maximum iterati
ons (200) reached and the optimization hasn't converged yet.
  warnings.warn(
```

Out[12]:
```
▾ MLPClassifier
MLPClassifier(hidden_layer_sizes=(30, 30, 30))
```

In [13]:
```
predictions = mlp.predict(X_test)
```

In [14]:
```
from sklearn.metrics import classification_report,confusion_matrix
print(confusion_matrix(y_test,predictions))
```

```
[[40  7]
 [11 85]]
```

In [15]:
```
print(classification_report(y_test,predictions))
```

```
              precision    recall  f1-score   support

           0       0.78      0.85      0.82        47
           1       0.92      0.89      0.90        96

    accuracy                           0.87       143
   macro avg       0.85      0.87      0.86       143
weighted avg       0.88      0.87      0.88       143
```

In [16]:
```
len(mlp.coefs_)
```

Out[16]:
```
4
```

In [17]:
```
len(mlp.coefs_[0])
```

Out[17]:
```
30
```

In [18]:
```
len(mlp.intercepts_[0])
```

Out[18]:
```
30
```

In [ ]:
```

```